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Module 2:

Energy and power

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1 Introduction

This chapter covers the concepts of energy, electrical work, power and efficiency. It is important to start by making a clear distinction between energy and work, as these two quantities are closely related and even share the same unit, but describe different things.

Learning objectives: Energy and electrical power

The students

- can name energy, electrical work, power and efficiency.
- understand how energy is converted and used in electrical systems.
- can calculate electrical work, power and efficiency.
- can analyse the efficiency of energy conversion systems.

The different variables are clearly distinguished from one another using the example of a pumped storage power station (1.1). A pumped storage power plant serves to convert electrical energy into potential energy and vice versa. As soon as the electrical energy demand of consumers is lower than the available supply, the electric motor of the pumped storage power plant pumps water from a lower to a higher level. In this process, the electric machine is operated as a (pump) motor. Via the magnetic field generated by an electric current in the motor winding, this produces a rotary motion which is used to pump the water into the upper reservoir. The electric motor thus converts electrical energy (the current flowing through the motor winding) into mechanical work (the water is pumped into the upper reservoir). Once the water has reached the upper reservoir, it has a higher potential than before in the lower reservoir. The water in the upper basin therefore stores potential energy and, in the case of a loss-free pump driven by a loss-free electric motor, this is exactly the same amount as was supplied to the electrical system as electrical energy. This is because the energy within a closed system must remain constant. In physics, this fundamental property is known as energy conservation.

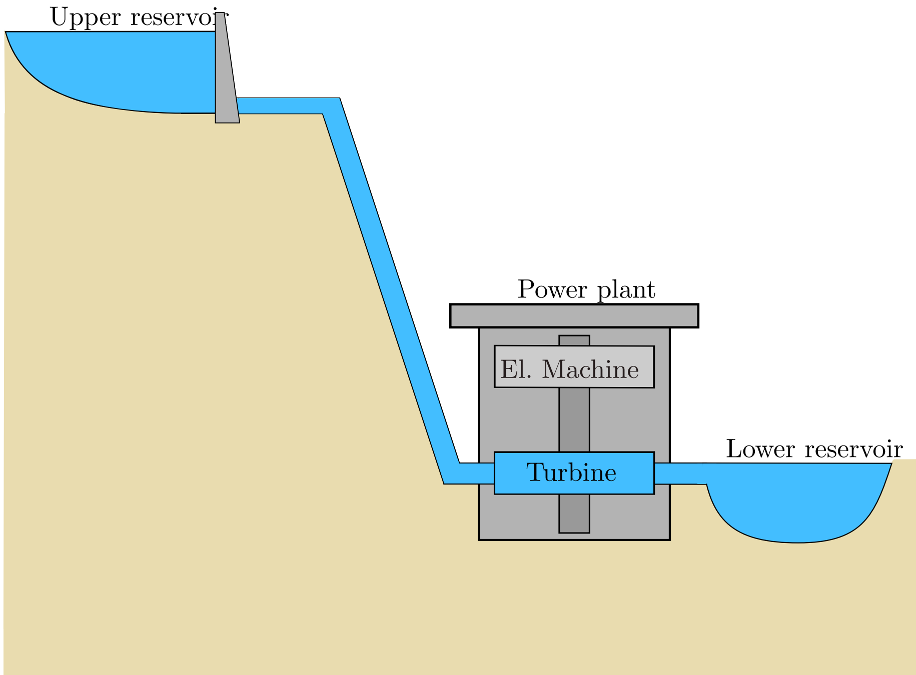

Figure 1.1: **Pumped storage power station.** The structure consists of an upper reservoir, a lower reservoir, turbines and an electric machine. It is used to store energy by pumping water uphill and to recover it by releasing hydropower to generate electricity.

As soon as the demand for electrical energy exceeds the supply, the water from the upper reservoir is discharged via a turbine. The turbine converts the previously stored potential energy of the water in the upper reservoir into mechanical work. This work performs a rotary motion in the electric machine. Since the machine now functions as a generator, it converts the mechanical work performed on it into electrical energy. During this process, a sluice valve in the upper reservoir can be used to reduce the amount of water flowing through the turbine within a certain period of time. This means that less potential energy is converted into mechanical work and thus electrical energy per unit of time. The work performed per unit of time is referred to as power. The power of a system therefore indicates how much work can be performed per unit of time. It is an important parameter for electrical systems. Within the electrical machine, part of the energy supplied is converted into heat in the line resistance, which is released into the environment. Furthermore, friction losses also occur in the mechanical elements (pump, turbine) and on the walls of the pipes, which is ultimately also released into the environment as heat. Consequently, the amount of electrical energy supplied to the system is always greater than the amount of electrical energy that can be extracted from the system. The ratio of the energy extracted to the energy supplied is described as efficiency.

The terms energy, power, electrical work and efficiency are explained in more detail below. For a better understanding, the basics from the previous chapter on charge, electrical potential and electrical current are assumed here.

Key point:

Energy can be stored, work is done.

2 The Energy

The stored work performance is referred to as energy according to the work expended. For example, when an object is heated, this stored heat is present as thermal energy. When the energy is released into the environment, work is performed again. Energy is therefore the ability of an object to perform work. As described above, it is a conserved quantity, which means that energy can neither be created nor destroyed. This chapter deals in detail with the concepts of energy conservation, the idea of different forms of energy and methods of storing it.

2.1 The law of conservation of energy

The law of conservation of energy states that the total amount of energy in a closed system remains constant, even though energy can be converted between different forms. This means that energy can neither be created nor destroyed, but can only be transferred from one form to another.

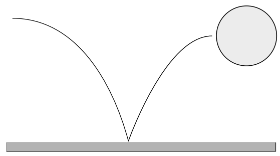

Figure 2.1: **Law of conservation of energy.** The potential energy of the ball is converted into kinetic energy, while the total energy content remains constant.

A clear example: when a ball is dropped from a height, its potential energy (the energy it has due to its position) is converted into kinetic energy (the energy of motion) as it falls. When it reaches the ground, some of this kinetic energy may be converted into other forms of energy, such as heat or sound, but the sum of all these forms of energy remains equal to the ball's original potential energy. This illustrates how energy is conserved even when it changes form.

Key point:

Energy is a conserved quantity.
Energy can neither be created nor destroyed.
Energy can only change its form.

Learning objectives: The Electric Charge

The students can

- describe the properties of electric charges and related physical phenomena
- describe electric fields and calculate them for simple charge arrangements
- Calculate forces on charges using Coulomb's law

2.2 The different forms of energy

In physics, energy describes the ability to perform work and occurs in various forms, each with specific properties and applications.

Kinetic energy is determined by the movement of an object. Energy depends on the mass and velocity of the object and is important in mechanics. **Potential energy** refers to the position of an object in a force field, such as the Earth's gravitational field. A ball that is thrown into the air transforms its potential energy into kinetic energy as it falls.x **Thermal energy**, oder Wärme, ist die Energie, die aus der Bewegung der Teilchen eines Objekts resultiert. Sie ist in Wärmekraftmaschinen und anderen thermodynamischen Prozessen von entscheidender Bedeutung. **Electrical energy** entsteht durch die Bewegung von Elektronen in einem elektrischen Feld und ist fundamental für den Betrieb technologischer Geräte. **Chemical energy** ist in den Bindungen von Molekülen gespeichert und wird bei chemischen Reaktionen freigesetzt. Sie wird zur Bereitstellung von nutzbarer Energie und in der Medizin genutzt. **Nuclear energy** entsteht durch Veränderungen im Atomkern, zum Beispiel durch Kernspaltung oder -fusion. Sie wird zur Bereitstellung von nutzbarer Energie und in der Medizin genutzt.

Key point: The forms of energy

Kinetic energy (energy of motion)
Potential energy (energy of position)
Thermal energy (heat energy)
Electrical energy
Chemical energy
Nuclear energy
Radiant energy

2.3 Electrical energy

Energy is the potential to perform work, for example, to accelerate charge carriers for a certain period of time. An energy source can only release as much energy as has been supplied to it beforehand. This corresponds to the time during which work was performed, i.e. an energy exchange took place. Since energy cannot exist without prior work and work cannot be performed without energy, causality is complex and mutually dependent. For this reason, the following assumes the existence of an energy source that can then perform work.

Energy E is generally defined over the integrated distance between two points P_1 and P_2 via a force $\vec{F}$.

$$E = \int_{P_1}^{P_2} \vec{F} \cdot \mathrm{d}\vec{s} \qquad [E] = 1 \text{ Joule} = 1 \text{ J} = 1 \text{ Nm} \tag{2.1}$$

An electric force $\vec{F}_{\mathrm{el}}$ is present when a charge Q is exposed to an electric field $\vec{E}$. Multiplying both quantities yields the following expression:

$$\vec{F}_{\mathrm{el}} = Q \cdot \vec{E} \tag{2.2}$$

Inserted into the formula 2.1, integration over the electric field strength $\vec{E}$ yields the electric energy E_{el}. Since the amount of charge Q does not depend on the distance, it can be moved outside the integral.

$$E_{\mathrm{el}} = Q \cdot \int_{P_1}^{P_2} \vec{E} \cdot \mathrm{d}\vec{s} \tag{2.3}$$

From Module 1, we know that the integral over an electric field under constant conditions is the voltage U. This results in the following simplified representation.

$$[E_{\mathrm{el}}] = 1 \text{ Joule} = 1 \text{ J} = 1 \text{ Ws}$$

$$E_{\mathrm{el}} = Q \cdot U \tag{2.4}$$

2.4 Electrical energy storage

Electrical energy storage devices are characterised by the fact that they store energy in electrical or magnetic fields. The basic storage devices of this type are capacitors and coils. The advantages of these energy storage devices are fast energy release and high efficiency. In practice, however, they have limited storage capacity and are expensive. Capacitors can undergo many charging cycles, but have a low energy density. In order to store energy in a coil, current must flow through it continuously, which is technically challenging and only possible in special cases..

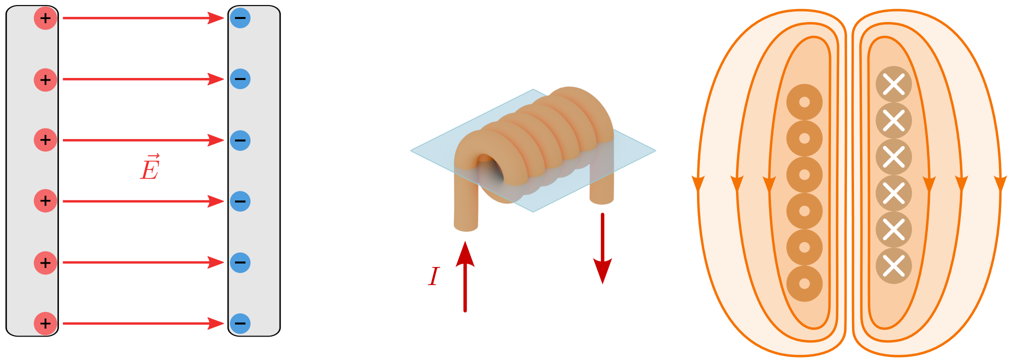

Figure 2.2: **Electricity storage systems.** An ideal plate capacitor (left) stores energy in an electric field, while an ideal coil (right) stores energy in a magnetic field.

Figure 2.2 shows the two basic types of electric energy storage. On the left, energy is stored in an electric field. This is achieved by placing charges with different signs on the capacitor plates. On the right-hand side, a coil is shown which, according to Lenz's law, generates a magnetic field due to a current flow. The topics of coils and capacitors are covered in more detail in Module 3. Since a current flow is required to maintain the magnetic field, a coil does not constitute a storage device in which electrical energy can be stored outside a circuit through which current flows. A capacitor, on the other hand, can store electrical energy for a long time even without a connected circuit. This means that a capacitor can continue to release energy long after an electrical circuit has been switched off. For example, the rear light of a bicycle can continue to be supplied with electrical energy even when the dynamo is stationary at a red traffic light. This effect is undesirable in circuits with higher voltages, for example, as the electrical energy present in the capacitors can pose a hazard to people even after the systems have been switched off if there are no appropriate protective circuits.

Batteries are therefore often used for medium-term storage of larger amounts of energy (see Figure 2.3). Batteries store energy chemically, can store larger amounts over longer periods of time and are versatile, making them ideal for powering mobile applications.

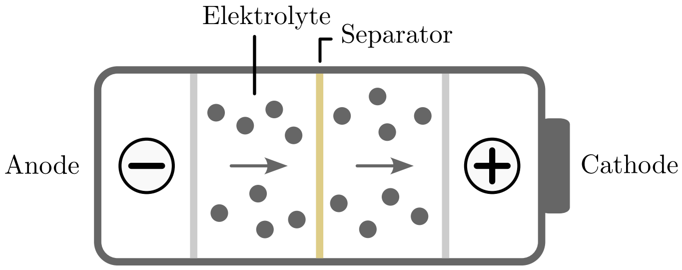

Figure 2.3: **Chemical energy storage.** Simple construction of a battery.

Although batteries are versatile, they also have some disadvantages: they are often expensive to purchase and maintain, have a limited lifespan, and their manufacture and disposal can cause significant environmental pollution due to the raw materials used.

3 The work

In this chapter, the symbols E for energy (without a vector arrow) and W for work are used to distinguish between work and energy. Although the symbol W is often used colloquially for both work and energy, work is essentially a change in energy. Both quantities, work and energy, are measured in joules (J). As described in the section on energy conservation, the law of conservation of energy states that energy can neither be created nor destroyed. It merely changes its form. This association is described by the following formula:

$$W = \Delta E \qquad [W] = 1 \text{ Joule} = 1 \text{ J} = 1 \text{ Nm} \tag{3.1}$$

This change in energy can be caused by various physical actions, including moving an object in a force field. To further illustrate these concepts, the general formula 3.2 for work is shown below:

$$W = -\int_{P_1}^{P_2} \vec{F} \cdot \mathrm{d}\vec{s} \tag{3.2}$$

The equation calculates work by integrating force along a path, whereby the direction of the force relative to the direction of movement of the object is crucial. A change of sign in the formula indicates that energy is being used from an energy store for work. This sign is context-dependent and changes depending on the perspective. If energy is withdrawn, the work is considered negative as it is subtracted from the total stored energy, signalling a reduction in the system's energy.

Key point:

The sign of the work is context-dependent: Work is positive when energy is added to a system and negative when energy is removed from a system.

3.1 Electrical work

In electrostatics, the forces $\vec{F}_{\mathrm{el}}$ acting between charge carriers Q and electric fields $\vec{E}$ are central. These forces arise when electric fields act on charge carriers. The electric field $\vec{E}$ is defined as the force per charge and describes how large the force $\vec{F}_{\mathrm{el}}$ is in relation to the applied charge Q.

$$\vec{E} = \frac{\vec{F}_{\mathrm{el}}}{Q} \quad \Rightarrow \quad \vec{F}_{\mathrm{el}} = Q \cdot \vec{E} \tag{3.3}$$

If we now substitute the formula for the electric force $\vec{F}_{\mathrm{el}}$ into the formula 3.2 for general work, we obtain the electric work W_{el}. Since the charge Q does not change over the distance s, it can be moved outside the integral. Since the electric field $\vec{E}$ is a vector and therefore has a magnitude and a direction, the product of $\vec{E}$ and $\mathrm{d}\vec{s}$ always depends on the chosen distance s. Therefore, a line integral must be formed using the product $\vec{E}$ and $\mathrm{d}\vec{s}$.

$$W_{\mathrm{el}} = \int_{P_1}^{P_2} -Q \cdot \vec{E} \cdot \mathrm{d}\vec{s} \quad \Rightarrow \quad W_{\mathrm{el}} = -Q \int_{P_1}^{P_2} \vec{E} \cdot \mathrm{d}\vec{s} \tag{3.4}$$

From Module 1, we know that the integral over $\vec{E} \cdot \mathrm{d}\vec{s}$ in the constant case yields the voltage U. For a consumer-independent representation, the negative sign is omitted in the following formula and simplified for the constant case by inserting the voltage U.

$$W_{\mathrm{el}} = Q \int_{P_1}^{P_2} \vec{E} \cdot \mathrm{d}\vec{s} \quad \Rightarrow \quad W_{\mathrm{el}} = Q \cdot U \tag{3.5}$$

Since an electric current is necessary for the transport of electrical energy, the relationship between electric current I and charge Q known from Module 1 can now be used and rearranged to replace the charge Q in formula 3.5. A constant current flow I results from a uniform movement of the charge carriers Q over time t. In this case, the differential operator (d) can be omitted. Rearranging for Q yields the following formula:

$$I = \frac{\mathrm{d}Q}{\mathrm{d}t} \quad \Rightarrow \quad Q = I \cdot t \tag{3.6}$$

Applied, this now results in the following expression, which illustrates that performing or applying work requires a flow of current.

$$W_{\mathrm{el}} = U \cdot I \cdot t \qquad [W_{\mathrm{el}}] = 1 \text{ Joule} = 1 \text{ J} = 1 \text{ Ws} \tag{3.7}$$

Key point:

- A certain amount of time is always required to complete a task.
- A momentary state is always the actual state of energy distribution.
- Performing electrical work always requires a flow of current.

3.2 Path integral of electrical work

The movement of a charge in an electric field depends solely on the change in position along the electric field lines. The corresponding sign is determined by the direction of movement along or against the field lines. Movements along an equipotential surface (surface with identical electric potential) have no influence on the electric work. In Figure 3.1, points P_1 and P_2 are shown at different potentials, which are connected by two different distances s_1 and s_2.

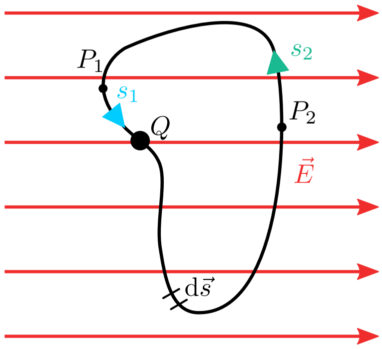

Figure 3.1: **Movement of a point charge.** Shows the movement of a point charge along two paths. The work done along a closed path in a static electric field is zero.

The magnitudes of two equidistant position changes in opposite directions along the electric field lines are identical. These properties mean that the change in position of a charge Q from point P_1 to point P_2 is equal in magnitude to any other distance from P_2 to P_1.

$$|W_{\mathrm{el}}| = \left| -Q \cdot \int\limits_{s_1} \vec{E} \cdot \mathrm{d}\vec{s} \right| = \left| -Q \cdot \int\limits_{s_2} \vec{E} \cdot \mathrm{d}\vec{s} \right| \tag{3.8}$$

This behaviour reflects the fact that the work is the same, but in one case work is expended and in the other case work is done. This leads to opposite signs, which means that the sum of both works cancels out. Mathematically, this can also be expressed by saying that the ring integral along any contour back to the starting point is zero. Since the charge Q is outside the circulation integral, the result is therefore independent of the amount of charge.

$$W_{\mathrm{el}} = -Q \cdot \int\limits_{s_1} \vec{E} \cdot \mathrm{d}\vec{s} - Q \cdot \int\limits_{s_2} \vec{E} \cdot \mathrm{d}\vec{s} = -Q \cdot \oint\limits_{s} \vec{E} \cdot \mathrm{d}\vec{s} = 0 \rightarrow \oint\limits_{s} \vec{E} \cdot \mathrm{d}\vec{s} = 0 \tag{3.9}$$

The result of the ring integral illustrates a fundamental principle of physics: in a static field, the work done over a closed path is always zero. This confirms the path independence of the work in these fields and shows that the work done depends exclusively on the starting and ending points and not on the specific path between these points.

Key point:

The line integral of a charge in a static electric field is always zero, regardless of the amount of charge.

4 Power

Power indicates how quickly work is performed or energy is transferred. It is defined as the amount of energy converted per unit of time. The SI unit of power is the watt (W), where one watt corresponds to one joule per second. In other words, the work described above is the integral of power over time.

$$W = \int P \cdot \mathrm{d}t \tag{4.1}$$

By rearranging the formula, we see that power P is the change in work over time.

$$P = \frac{\mathrm{d}W}{\mathrm{d}t} \tag{4.2}$$

4.1 The electrical power

In the case of electrical power, assuming constant work over time after the onset of electrical work W_{el}, the following expression results from formula 3.7:

$$P = \frac{W_{\mathrm{el}}}{t} = \frac{U \cdot I \cdot \cancel{t}}{\cancel{t}} = U \cdot I \qquad [P] = 1\ \text{Watt} = 1\ \mathrm{W} \tag{4.3}$$

4.2 The different power types

In addition to electrical power, there are other power quantities that result from the various forms of energy. Furthermore to the forms of energy discussed in section 2.2, the corresponding power quantities are listed in the following table. In each case, the potential quantity is multiplied by the flux quantity. The table only applies to direct current.

Types	**Potential value**	**Flow rate**	**Formula**
Electric	Voltage (U)	Current (I)	$P_{\mathrm{el}} = U \cdot I$
Translatorical	Force (F)	Velocity (v)	$P_{\mathrm{tr}} = F \cdot v$
Rotational	Moment (M)	Angular velocity (ω)	$P_{\mathrm{rot}} = M \cdot \omega$
Thermal	Temperature diff. (ΔT)	Heat transfer ($k \cdot A$)	$P_{\mathrm{th}} = \Delta T \cdot k \cdot A$
Fluidic	Pressure (p)	Volume flow ($\dot{V}$)	$P_{\mathrm{fl}} = p \cdot \dot{V}$

Table 4.1: **Types of performance.** Shows the different types of power in different physical systems.

Key point:

Electrical work measures the energy transferred, while electrical power indicates the transfer speed.

The different types of power are relevant to the efficiency of a system. As a rule, in addition to the desired power, undesirable power also occurs, which should be avoided if possible. This is discussed in the following section on efficiency.

5 The efficiency

Efficiency describes the ratio of output power (useful power) to input power (energy supplied) in a system and is a measure of the efficiency with which a system converts energy. Using the example of an electric motor, Figure 5.1 shows how its power balance is composed.

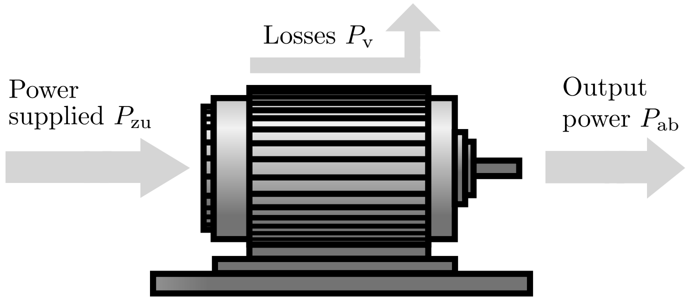

Figure 5.1: **Efficiency of an electric motor.** Shows the ratio of output power to input power. Efficiency describes the efficiency of the motor in converting energy.

The electric motor is supplied with electrical energy at the input, which it absorbs as input power P_zu. The desired output power P_ab in this case is of a rotary nature. The waste heat generated by the motor, as well as the noise and friction it causes, together result in the power loss P_v.

5.1 Calculation of efficiency

Efficiency is typically expressed as a percentage. An efficiency of 100means that all of the energy supplied is effectively converted into useful energy. However, this is not achievable in practice due to losses, usually in the form of heat, friction or sound. Mathematically, efficiency η (eta) is defined as follows:

$$\eta = \frac{P_\text{out}}{P_\text{in}} \cdot 100\% \tag{5.1}$$

In reality, efficiency will never be 100To keep losses to a minimum, it is important to be aware of the mechanisms that cause them and to reduce them wherever possible. The following section explains the principle of loss mechanisms using a light bulb as an example.

5.2 Loss mechanisms of a light source

A prime example of the relevance of efficiency is the classic incandescent light bulb. It produces light by passing electricity through a wire, which heats up so much that it begins to glow. Less than 5 per cent of the energy used is converted into light. More than 95% of the energy is converted into heat. Due to their poor efficiency, incandescent light bulbs have been gradually banned in the EU since 2009. Figure 5.2 shows a classic incandescent light bulb on the left and an energy-saving light bulb on the right.

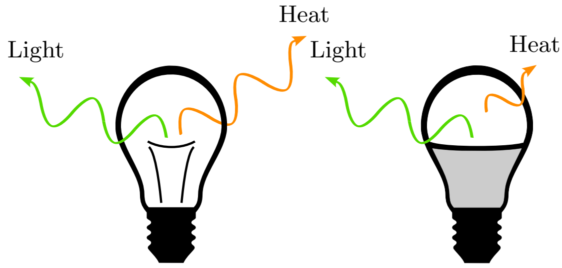

Figure 5.2: **Energy conversion in incandescent bulbs and energy-saving bulbs.** Whereas an incandescent bulb converts most of the electrical energy it consumes into heat and only a small portion into light, an energy-saving bulb converts a significantly higher proportion of the electrical energy into light.

High efficiency is desirable in many applications, as it means that less energy is converted into forms of energy that are generally undesirable, and instead remains available to the system as useful energy. In practice, however, many factors such as material properties, design and operating conditions are crucial for maximising efficiency. Efficiency plays a crucial role in environmental and energy technology, as it is directly related to energy consumption and environmental impact. More efficient systems can help to reduce energy demand and greenhouse gas emissions.

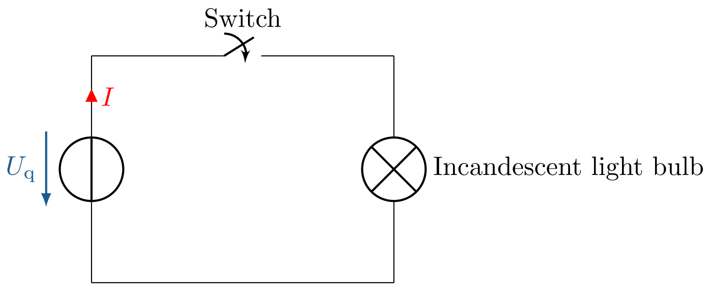

Figure 5.3: Example circuit Efficiency

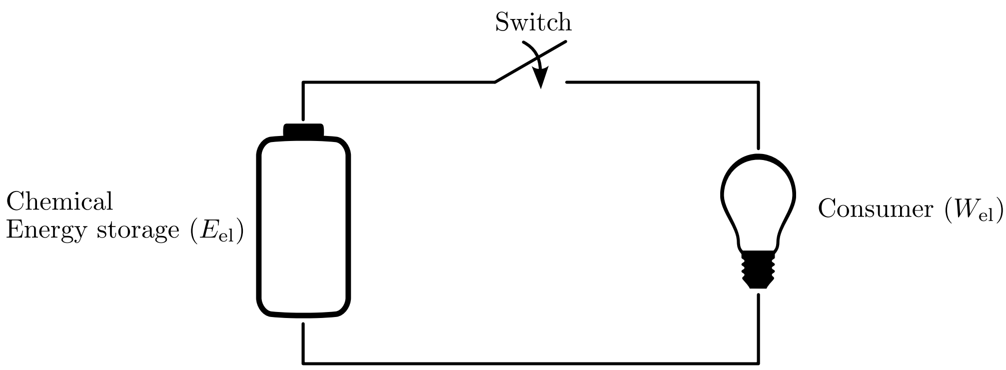

Figure 5.4: **Simple circuit.** shows a simple electrical circuit consisting of a battery, a switch and a light bulb. Closing the switch completes the circuit, causing the light bulb to glow by converting electrical energy into light and heat.

Example 5.1: Calculation of efficiency

A heat engine extracts energy of $Q_{\text{zu}} = 5000\,\text{J}$ from a heat source and performs mechanical work of $W = 1200\,\text{J}$ in the process..

Wanted: Efficiency η

Formula:

$$\eta = \frac{\text{Useful work}}{\text{Supply of thermal energy}} \cdot 100\% = \frac{W}{Q_{\text{zu}}} \cdot 100\%$$

Insert:

$$\eta = \frac{1200\,\text{J}}{5000\,\text{J}} \cdot 100\% = 24\%$$

The efficiency is $\eta = 24\%$.

A Exercise tasks

A.1 Work and energy

An electric motor lifts a mass of m=10 kg to a height of 5 m.

a) Calculate the compressed work.

b) How much energy is required if the efficiency of the motor is $\eta = 80\%$?

A.2 Potential energy in the reservoir

A reservoir contains 2,000,000m^3 of water at a height of 100m. Calculate the potential energy stored. $\rho = 1000\,\mathrm{kg/m^3}, \quad g = 9{,}81\,\mathrm{m/s^2}$

A.3 Kinetic energy of a car

A car with a mass of 1200 kg is travelling at 72 km/h. Calculate its kinetic energy.

A.4 Electrical energy of a device

A 2000 W kettle runs for 3 minutes.

a) Calculate how much electrical energy it has consumed.

b) How much energy is needed to heat 5 litres of water from 20 °C to 100 °C?

A.5 Energy and power

A heating device has a power consumption of P = 2 kW and is operated for t = 3 hours.

a) How much energy is consumed?

b) If the electricity price is €0.30/kWh, how much will it cost?

A.6 Efficiency

A generator produces an electrical power of P_{el} = 1 kW by absorbing a mechanical power of P_{mech} = 1.5 kW.

a) Calculate the efficiency

b) How large must P_{mech} be to generate P_{el} = 2 kW?

B Solutions to the exercises

B.1 Work and energy

a)

$$W = m \cdot g \cdot h$$

$$W = 10kg \cdot 9{,}81\frac{m}{s^2} \cdot 5m$$
$$W = 490{,}5J$$

b)

$$Efficiency = \frac{Energy\ consumed}{Energy\ expended} = \eta$$
$$E_{zu} = \frac{W}{\eta} = \frac{490{,}5J}{80\%} = 613{,}125J$$

B.2 Potential energy in the reservoir

a)

$$E_{\text{pot}} = m \cdot g \cdot h$$
$$m = \rho \cdot V = 1000\,\text{kg/m}^3 \times 2\,000\,000\,\text{m}^3 = 2\,000\,000\,000\,\text{kg}$$
$$E_{\text{pot}} = 2\,000\,000\,000\,\text{kg} \times 9{,}81\,\text{m/s}^2 \times 100\,\text{m}$$
$$E_{\text{pot}} = 1{,}962 \times 10^{12}\,\text{J}$$

B.3 Kinetic energy of a car

a)

$$E_{\text{kin}} = \frac{1}{2}mv^2$$
$$m = 1200\,\text{kg}$$
$$v = 72\,\text{km/h} = 72 \times \frac{1000\,\text{m}}{3600\,\text{s}} = 20\,\text{m/s}$$
$$E_{\text{kin}} = \frac{1}{2} \times 1200\,\text{kg} \times (20\,\text{m/s})^2$$
$$E_{\text{kin}} = 240\,000\,\text{J}$$

B.4 Electrical energy of a device

a)

$$P = 2000\,\text{W}$$
$$t = 3\,\text{min} = 180\,\text{s}$$
$$E = P \cdot t = 2000\,\text{W} \times 180\,\text{s} = 360\,000\,\text{J}$$

b)

$$m = 5\,\text{kg} \quad \text{(Density of water} \approx 1\,\text{kg/L)}$$
$$c = 4186\,\frac{\text{J}}{\text{kg} \cdot \text{K}} \quad \text{(Specific heat capacity of water)}$$
$$\Delta T = 100°\text{C} - 20°\text{C} = 80\,\text{K}$$
$$Q = m \cdot c \cdot \Delta T = 5\,\text{kg} \times 4186\,\frac{\text{J}}{\text{kg} \cdot \text{K}} \times 80\,\text{K} = 1\,674\,400\,\text{J}$$

B.5 Energy and power

a)

$$E = P \cdot t$$

$$E = 2\,\mathrm{kW} \cdot 3\,\mathrm{h} = 6\,\mathrm{kWh}$$

b)

$$C = E \cdot P$$

$$C = 6\,\mathrm{kWh} \cdot 0{,}30\,\frac{\mathrm{Euro}}{\mathrm{kWh}} = 1{,}80\,\mathrm{Euro}$$

B.6 Efficiency

a)

$$\eta = \frac{P_{\mathrm{el}}}{P_{\mathrm{mech}}} = \frac{1\,\mathrm{kW}}{1{,}5\,\mathrm{kW}} \approx 0{,}6667 \Rightarrow \eta \approx 66.67\%$$

b)

$$P_{\mathrm{mech}} = \frac{2\,\mathrm{kW}}{0{,}6667} \approx 3\,\mathrm{kW}$$